



Phishing Awareness Training

Cyber Security Internship Project

Prepared for students and beginner cybersecurity learners — concise, practical guidance to recognize and respond to phishing threats.

- ⓘ Speaker notes: Welcome attendees. Briefly introduce the internship project, goals of the workshop, and expected learning outcomes: recognize phishing, analyze examples, and practice safe responses.

Introduction to Phishing

Definition: Phishing is a social-engineering attack that uses deceptive communications (usually email, SMS, or calls) to trick people into revealing sensitive information or installing malware.

Why it's common: Low technical cost for attackers, high human success rate, and scalable distribution make phishing the most prevalent cyber threat.

- Targets credentials, financial data, or internal systems
- Often leverages urgency and trust
- Easy to automate and personalize

Speaker notes: Emphasize how phishing combines simple tricks with psychology — attackers rely on human behavior more than complex hacking.



Visual: Attack distribution and human targets

Why Phishing Is Dangerous



Data Breach Risk

Compromised credentials can expose personal and corporate data, leading to large-scale breaches.



Financial Impact

Direct fraud, wire-transfer scams, and remediation costs can be substantial for victims and organizations.



Reputational Damage

Customers and partners lose trust after a successful phishing incident, harming long-term relationships.



Secondary Attacks

Phishing is often the entry vector for malware, ransomware, and Business Email Compromise (BEC).

Statistic: Recent industry reports show phishing is involved in a large percentage of breaches — human risk remains the weakest link.



Speaker notes: Provide a concrete example of a phishing-caused breach in the news (keep it brief). Emphasize impacts beyond immediate loss: downtime, legal exposure.

Types of Phishing Attacks



Email Phishing

Mass emails that impersonate trusted senders to collect credentials or deliver malware.



Spear Phishing

Targeted messages crafted using personal information to increase credibility and success rate.



Whaling

High-value targeting of executives or finance officers to execute fraud or data theft.



Smishing

Phishing via SMS — often includes short links and urgent language to prompt action on mobile.



Vishing

Voice phishing over phone calls — attackers impersonate banks, IT, or other authorities to extract information.

❏ Speaker notes: Clarify distinctions and risk levels. Ask learners to consider which types they encounter most in daily life.

Anatomy of a Phishing Email



This diagram outlines the typical steps an attacker follows and how a single click can escalate into credential theft or system compromise.

 Speaker notes: Walk through each step slowly. Explain spoofing techniques (display name vs. actual address) and why attachments are risky.

Common Red Flags in Emails

Suspicious Sender Address

Display name mismatch, irregular domain, or subtle typos (ex: company-support.com vs company.com).

Urgent or Threatening Language

Pressure to act immediately (account suspension, payment due) is a common manipulation tactic.

Unexpected Attachments

Unknown .zip, .exe, or Office files with macros — do not open without verification.

Fake or Masked Links

Hover to preview links — mismatched URLs or domain misspellings often reveal phishing.

Poor Grammar & Generic Greetings

Spelling errors, unusual phrasing, or generic salutations (Dear Customer) can indicate fraud.

 Speaker notes: Demonstrate how to inspect headers and hover-over links. Encourage students to ask IT before interacting with suspicious emails.

Fake Websites & URL Spoofing

Attackers create convincing fake sites to capture credentials. Common tricks include:

- Subdomain tricks (login.example.com.attacker.com)
- Character substitution (ex: using 'rn' for 'm')
- HTTPS padlock misuse — presence of HTTPS alone $\neq$ trustworthiness
- URL shorteners hiding final destination

How to verify: Manually type known URLs, check certificate details, and inspect full domain in the address bar.

- ❏ Speaker notes: Show a live demo or screenshots of subtle URL differences. Emphasize manual typing for sensitive logins.



Visual: Example of URL manipulation

Social Engineering Techniques Used by Attackers



Authority

Impersonating executives, IT, or vendors to compel compliance without verification.



Urgency & Scarcity

Creates panic or fear to bypass rational checks (e.g., “Pay now or account closed”).



Trust & Familiarity

Using known logos, writing style, or previously compromised accounts to seem legitimate.



Curiosity & Incentives

Promising rewards, invoices, or urgent documents to entice clicks.



Speaker notes: Use role-play exercises — have learners craft verification questions to ask a supposed requester before sharing sensitive info.

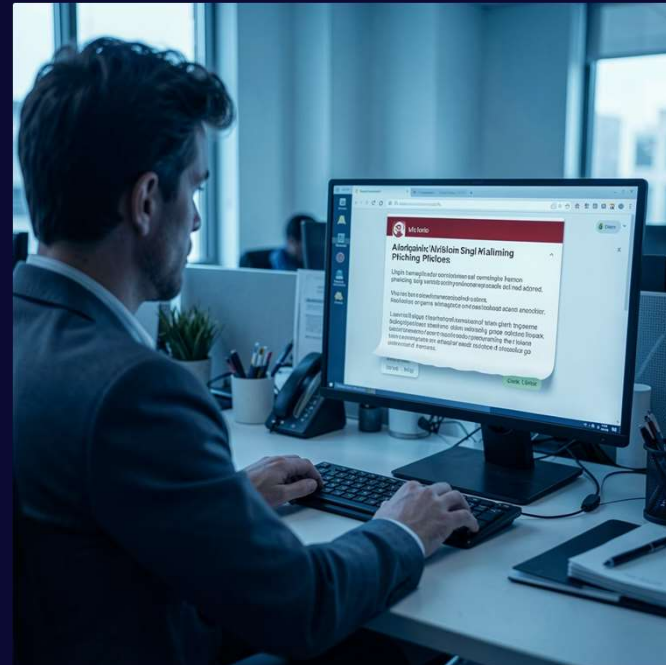
Real-World Phishing Case Studies

Case Study 1 — Business Email Compromise (BEC)

An attacker impersonated a CFO and requested an urgent wire transfer. Lack of secondary verification led to loss of funds. Lessons: verify payment changes via known phone numbers; require dual approval for transfers.

Case Study 2 — Credential Harvesting

Mass email with a fake corporate portal collected employee credentials. Attackers pivoted to internal systems. Lessons: enforce MFA and monitor for anomalous logins.



Visual: Realistic scenario for discussion

- ① Speaker notes: Present timelines of each incident, list immediate remediation steps taken, and discuss longer-term controls implemented by the organization.

Best Practices to Avoid Phishing Attacks

Verify Sender Identity

Confirm unusual requests through an independent channel (phone or internal chat).

Use Multi-Factor Authentication

MFA reduces risk even if credentials are compromised.

Avoid Clicking Suspicious Links

Hover to preview and type URLs manually for sensitive sites.

Keep Software Updated

Patch systems and enable secure email filtering to reduce exposure to exploits.

Report Suspicious Activity

Use your organization's reporting channel — quick reporting helps limit damage.

Quick checklist: Verify → Think → Confirm → Report. Adopt a skeptical mindset and follow team escalation procedures.

- ✔ Speaker notes: Conclude with actionable next steps: complete the interactive quiz, practice email inspections, and schedule a follow-up session for hands-on exercises. Encourage questions.

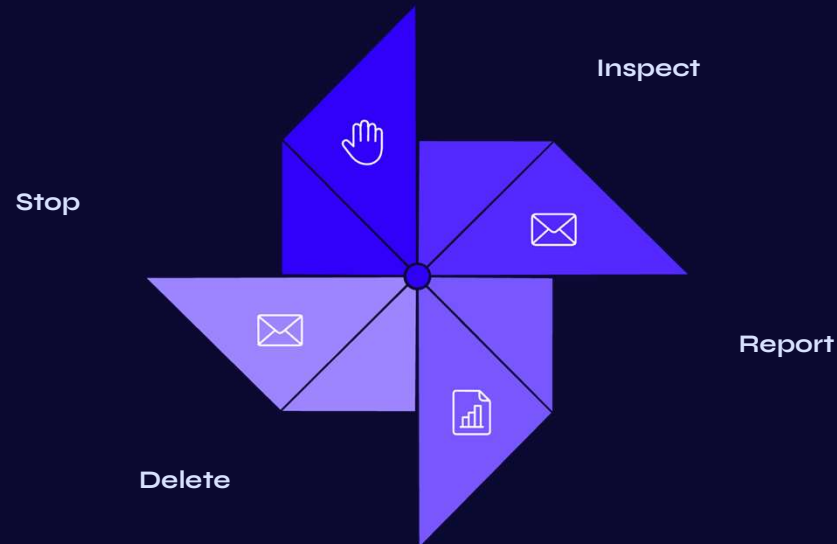
What to Do If You Suspect a Phishing Attempt

Phishing is the #1 cause of data breaches. Knowing how to respond quickly can protect you — and your organization.

- ❏ **Speaker Notes:** Open by asking the audience: "Has anyone ever received a suspicious email?" Use their responses to transition into the topic. Emphasize that phishing is the most common entry point for cyberattacks — it accounts for over 90% of successful breaches. Set the expectation that by the end of this session, everyone will know exactly what to do when they spot a suspicious message.



What to Do If You Suspect a Phishing Attempt



Acting quickly and calmly is key. Never engage with the sender — simply report and delete.

Red Flags to Watch For: Urgent language, mismatched URLs, unexpected attachments, and requests for credentials.

Speaker Notes: Walk through each step slowly. Pause on "Stop" — this is the most critical action. Ask the audience: "What's the first thing you'd do if you got a suspicious email?" Reinforce that they should never click "unsubscribe" on a phishing email, as it confirms their address is active. Mention that many organizations have a dedicated "Report Phishing" button in their email client.

Cybersecurity Tools for Protection



Antivirus & Anti-Malware

Scans files and blocks malicious software before it can do damage.



Multi-Factor Authentication

Adds a second layer of verification — even stolen passwords won't grant access.



Email Filtering

Automatically flags and quarantines suspicious messages before they reach your inbox.



VPN & Encryption

Secures your connection and scrambles data so it's unreadable to attackers.



Speaker Notes: Emphasize that tools are only effective when used correctly. MFA is the single most impactful step individuals can take — it blocks 99% of automated attacks. If time allows, do a quick live demo of enabling MFA on a common platform like Gmail. Remind the audience that no single tool is a silver bullet; layering defenses is what creates real security.

Interactive Quiz

Test your knowledge — no wrong answers, only learning opportunities!

1

What's the #1 sign of a phishing email?

A) Poor grammar B) Urgent request for credentials C) Unknown sender D) All of the above

2

What should you do first when you spot a suspicious email?

A) Reply to ask who they are B) Click the link to investigate C) Stop — don't interact D) Forward to a friend

3

What does MFA stand for?

A) Multi-Factor Authentication B) Maximum Firewall Access C) Malware Filter Alert D) Multi-File Authorization

4

Which tool blocks malicious software before it runs?

A) VPN B) Antivirus C) Email filter D) Password manager

5

True or False: You should click "unsubscribe" on suspicious emails.

A) True — it stops the emails B) False — it confirms your address is active



Speaker Notes: Go through each question one at a time and let the audience vote by raising hands or using a polling tool if available. Reveal answers after each question and briefly explain why. Answers: 1-D, 2-C, 3-A, 4-B, 5-B. Use this as a moment to reinforce key concepts from earlier slides. Keep energy high — this is a low-stakes, fun way to cement learning.

Key Takeaways



Stop First

Never click or reply — pause before you act.



Inspect Carefully

Check sender addresses, URLs, and language for red flags.



Report It

Alert your IT or security team immediately.



Layer Your Defenses

Use MFA, antivirus, and stay vigilant every day.

Thank You — Questions?



Speaker Notes: Recap the four key takeaways briefly. Open the floor for questions — if no one asks, prompt with: "What's one thing you'll do differently after today?" Share any internal reporting contacts or resources (IT helpdesk, security team email). End on an encouraging note: cybersecurity is a team sport, and every person who learns these habits makes the whole organization safer.